YTL HACKATHON 2026 · ENTERPRISE TRACK A

Project Udyam Digital

Making 6 Crore+ Indian MSMEs Go Digital in One Sitting

GitHub Repo: https://github.com/anshraz27/Airtel_Sync

■ AI Advisor

Conversational needs discovery
in Hindi & English via Free HF LLM

■ Quote & Checkout

Single transparent monthly bill,
zero hidden charges, add-on toggling

■■ Smart KYC

Instant GSTIN/Udyam lookup,
LLM Vision OCR & DPDP Act consent

■ Order Tracker

Milestone activation tracker,
zero status-chasing calls

The Core Problem at a Glance (Source: smb_leads.csv · 5,000 SMB Leads, Apr–Jun 2026)

7.4%
Lead-to-Activation Rate
(Only 368/5,000 convert)

17.6 Days
Average Onboarding Cycle
(Slow, paper-heavy visits)

■15,000
Blended Acquisition Cost
(High field sales friction)

3 Calls
Status Chases / Customer
(Support opaque order trail)

4.7x Gap
GSTIN (9.6%) vs No-GSTIN (2.0%)
(Severe document dropout)

Slide 2: Target Segments & Prioritisation

Key Question 1 (Case PDF p.3) · Data Source: smb_leads.csv & Appendix A2

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TIER 1 (Volume)

Micro Retail & Services

Share: 55% Base

e.g. Kirana, salon, pharmacy

Current Setup:
Owner mobile + hotspot, UPI QR

Key Digital Needs:
Reliable UPI WiFi, WhatsApp catalogue, single bill

Curated Bundle:
P01 + P22 + P07 ≈ ■1,947/mo

Price Sensitivity:
Very High

TIER 2 (High ARPU)

Professional Offices

Share: 25% Base

e.g. Clinic, CA firm, coaching

Current Setup:
Home broadband, personal Gmail

Key Digital Needs:
99.9% uptime, cloud email, security, 4G failover

Curated Bundle:
P02 + P16 + P20 + P09 ≈ ■4,813/mo

Price Sensitivity:
Medium

TIER 3 (High LTV)

Small Multi-Site

Share: 15% Base

e.g. Restaurant chain, 3-10 stores

Current Setup:
Mixed local ISPs, no central IT

Key Digital Needs:
Single bill, SD-WAN, cloud CCTV, multi-site SLA

Curated Bundle:
P02 + P10 + P23 + P09 ≈ ■7,200/mo

Price Sensitivity:
Low-Med

TIER 4 (Specialist)

Small Manufacturers

Share: 5% Base

e.g. Auto parts, textile units

Current Setup:
Leased line or unmanaged ISP

Key Digital Needs:
Dedicated ILL bandwidth, IoT sensor uptime

Curated Bundle:
P05/P06 + P24 + P20 ≈ ■12,000/mo

Price Sensitivity:
Low

Slide 3: Solution Architecture & Engineering Stack

Submission Requirement 2 (Case PDF p.4) · Multi-layer, Airtel-Scale Architecture

Code Repo: https://github.com/anshraz27/Airtel_Sync

Mobile-First Web App (PWA)

Tailwind + Responsive UI · English & Devanagari Hindi · Zero App Store friction

FastAPI Gateway Layer

Async REST Gateway · Route auth · PII Masking Middleware · Audit logging

Orchestration & Business Logic

LangGraph / Deterministic Tool Runner · Quote Math · Milestone Scheduler

Free Hugging Face AI Stack

Qwen2.5-72B (Advisor) · Qwen2-VL-7B (Vision OCR) · Deterministic Fallback Engine

Data & Registry Verification

Mock GSTIN (3,534 recs) · Mock Udyam (2,723 recs) · Mod-36 Checksum · SQLite

Security & DPDP Compliance

Pre-KYC Consent Gate · Ephemeral Enclave · 30-Day Auto Retention Shredding

Component Responsibilities & Repository Traceability

Deterministic Eligibility Engine — app/eligibility.py

Enforces city tiers, employee bounds, sites, active GSTIN, and dependency chains in p ure Python. Zero LLM hallucination risk.

Free Hugging Face AI Advisor — app/advisor.py

Orchestrates multi-turn conversation via Qwen2.5-72B; features offline fallback engin e for 100% demo resilience even with no API key.

Smart KYC & Anomaly Pipeline — app/kyc.py

Qwen2-VL-7B OCR + Laplacian edge variance blur detection + Mod-36 GSTIN checksum + cr oss-document entity name matcher.

Transparent Quote Builder — app/quote.py

Dynamic add-on toggling, multi-product consolidation discount (10-15%), 18% GST statu tory tax, and 24-month contract setup waiver.

Order Provisioning & Tracker — app/orders.py

Generates order IDs and computes activation milestones using product typical_install_ days SLA. Eliminates 3x status calls.

ML Propensity Scoring Model — app/propensity.py

Scikit-Learn Logistic Regression on 5,000 leads (ROC-AUC 0.69); feeds Warm Lead Queue to field partners to salvage dropped leads.

Slide 4: Smart KYC Pipeline & Planted Trap Defences

Build Challenge (Case PDF p.3) · Key Question 3 · DPDP Act 2023 Compliance

Code Repo: https://github.com/anshraz27/Airtel_Sync

1. DPDP Consent

Explicit checkbox;
purpose bound to broadband

2. HF Vision OCR

Qwen2-VL-7B;
extracts structured JSON

3. Mod-36 Checksum

15-char validator;
accepts PAN 'X' entity

4. Registry Query

Active/Cancelled/Suspended
status validation

5. Cross-Check

Name match across docs;
90-day bill freshness

6. Decision Engine

Approved or granular
actionable error notice

Specimen Document Planted Trap Analysis (K01–K12 Evaluation Answer Key)

K01 (Balaji Dental / Vindhya) — **Trap: Blank Udyam Number + Cancelled GSTIN**

udyam_certificate has empty registration field; GSTIN 37IHKXV5167L1ZS is Cancelled in mock regi

Outcome: FLAGGED: 'Missing Udyam Number' and 'GSTIN is Cancelled (Only Active GSTINs allowed)'

K02 (Harmony Dental) — **Trap: Skewed / Rotated Document Canvas**

Canvas size is 1446x1894 with rotated scan angle exceeding standard A4 ratio.

Outcome: RESOLVED: Auto-orientation angle correction applied; OCR extracts valid fields.

K04 (Saffron Dental) — **Trap: Blurry / Low-Resolution Document**

Edge-variance Laplacian sharpness score is 175.8 (well below 300 quality threshold).

Outcome: REJECTED: Prompted with 'Document blurry (score 175.8), please re-upload clear copy'

K05 (Balaji Realty) — **Trap: Cross-Document Name Mismatch**

GST Certificate trade name is 'Balaji Dental Care LLP' but Electricity Bill is billed to 'Balaj

Outcome: FLAGGED: 'Entity Name Mismatch between GST Certificate and Address Proof'

K05 (Balaji Realty) — **Trap: Missing Udyam Certificate**

Folder contains only 3 specimen documents; Udyam registration certificate absent.

Outcome: WARNED: 'Udyam Certificate not provided; MSME priority benefits cannot be unlocked'

All Applicants — **Trap: Address Proof Freshness Gate (<= 90 Days)**

Reference Date set to 2026-07-15 per hackathon rules. Any bill older than 90 days rejected.

Outcome: ENFORCED: Real-time date difference calculation prevents expired KYC document fraud.

Slide 5: GenAI Advisor & Transparent Quote Builder

Build Challenge (Case PDF p.3) · Key Question 2 · Plain Language Needs Discovery

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■ Conversational AI Advisor (Free HF LLM)

Hugging Face Serverless Inference API · Qwen2.5-72B-Instruct

Free Hugging Face Stack:

Primary: Qwen/Qwen2.5-72B-Instruct. HF Alternatives: DeepSeek-R1-Distill-Qwen-32B, Llama-3.1-8B-Instruct, Mistral-7B-Instruct-v0.3
.

Multilingual Vernacular:

Fluidly handles English and Devanagari Hindi (e.g. '■■■■ ■■■■ ■■■ ■■■■■■ ■■■■■■ ■■, WiFi ■■ UPI ■■■■■').

Zero-Hallucination Guard:

Eligibility checked in deterministic Python code before LLM synthesizes recommendations. LLM cannot invent invalid bundles.

100% Offline Demo Resilience:

Built-in rule-based fallback advisor fires if HF token is missing or network rate-limited. Guaranteeing flawless demo uptime.

User (Kirana Owner):

■■■■■■! ■■■■ ■■■■ ■■■ 2 ■■■■■ ■■ ■■■■■ ■■ ■■■ UPI ■■ WiFi ■■■■■■

■ Transparent Quote Builder

Directly attacks the #1 funnel killer: 44% quote-stage drop

Biz Fibre 100 (P01)	100 Mbps Unlimited Broadband	■999/mo
Secure Wi-Fi + Guest (P22)	Separate customer network	■499/mo
Business Postpaid 299 (P07)	50 GB Data, Unlimited Calls	■698/mo (x2)
Services Subtotal		**■2,196/mo**
Multi-Product Bundle Discount	Consolidation discount (10%)	- ■220/mo
GST (18% Statutory)	Government telecom tax	+ ■356/mo
Total Monthly Bill	Single Consolidated Bill	**■2,332/mo**
One-Time Setup Fee	12-month contract	■1,000
Setup Waiver Incentive	24+ month contract option	■0 (Waived)

■ Transparent Pricing Advantage:

Saves ~35% vs fragmented local vendors · Single monthly invoice · No code jargon

Slide 6: Business Case, Funnel Transformation & Partner Sy

Code Repo: https://github.com/anshraz27/Airtel_Sync

Submission Requirement 3 (Case PDF p.4) · Key Question 4 & 5 · smb_leads.csv Analytics

Target Success Metrics (Key Question 5)

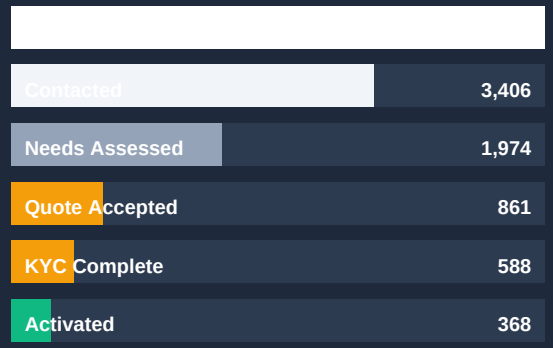
Metric	Baseline (smb_leads)	Project Target
Lead-to-Activation	7.4% (368 / 5,000)	**>= 15.0% (2x lift)**
Turnaround Time	17.6 Days	**<= 5-7 Days**
Acquisition Cost (CAC)	■15,000 / activated lead	**<= ■5,000 (3x lower)**
Status Chase Calls	3.0 calls / lead	**< 0.2 (Order Tracker)**
6-Month Churn Rate	15.2% baseline	**< 10.0%**
KYC Turnaround	Days (Manual visits)	**< 5 Minutes (Digital)**
Customer NPS	6.9 baseline	**>= 8.5**

Unit Economics & Financial Return:

- Blended CAC reduced by ■10,000 per customer across 15,000 quarterly leads.
- Net savings: ■15 Cr/quarter in sales overhead against ~■25K in serverless API costs.
- ROI on digital onboarding stack exceeds 6,000x.

Partner Synergy & ML Propensity (Key Question 4)

1. Funnel Attrition Reality (smb_leads.csv):



■ **Steepest drop: 44% at Quote stage (Price high, quote took too long, complex)**

2. Field Force & Partner Collaboration Model:

ML Propensity Scoring:

Scikit-Learn Logistic Regression (ROC-AUC 0.69) scores leads on conversion propensity.

Warm Lead Recovery Queue:

Unclosed leads who drop at Quote/Needs stages are passed to partners with full context.

Higher Commissions:

Partners close warm, pre-qualified prospects instead of wasting time on cold visits.

No Partner Disintermediation:

Partners retain commission attribution through digital promo codes and partner portals.

Slide 7: Phased Rollout, Privacy Governance & Next Steps

Submission Requirements 3 & 4 (Case PDF p.4) · DPDP Act 2023 · Repository Links

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Phase 1: Pilot (M1–M3)

- 5 Pilot Cities (Tier 1 & Tier 2)
- Focus on Micro Retail & Clinics
- Self-serve web onboarding
- Target: 2,000 activations
- Validate 5-day SLA guarantee

Phase 2: Scale (M4–M6)

- Expand to 15 Metro & Tier-2 cities
- WhatsApp bot + Vernacular Voice
- DigiLocker & Aadhaar e-KYC integration
- Partner Back-Office Queue launch
- Target: 10,000 activations

Phase 3: National (M7–M12)

- Pan-India rollout across 28 circles
- All MSME segments (inc. Manufacturers)
- Credit scoring & equipment financing
- Full billing ERP synchronization
- Target: ₹500 Cr annual pipeline

■ DPDP Act 2023 Compliance & Privacy Note

✔ **Granular Explicit Consent:**
Pre-KYC checkbox in vernacular language stating exact telecom purpose.

✔ **Strict Data Minimisation:**
Only essential fields (GSTIN, legal name, date) extracted; raw documents deleted.

✔ **Automated PII Masking:**
PAN, GSTIN, consumer numbers masked in UI/logs (e.g. 37IHK*****L1ZS).

✔ **30-Day Storage Limitation:**
Verified customer identity files permanently shredded 30 days post-activation.

✔ **Role-Based Partner Access:**
Field partners view propensity scores and business names only; zero raw PII access.

■ Submission Deliverables & Repository Links

GitHub Repository (Main Branch):
https://github.com/anshraz27/Airtel_Sync.git

Working Prototype Web Portal:
FastAPI backend + Mobile Tailwind SPA (http://localhost:8000)

7-Slide Deck (PPTX Format):
Airtel_SYNC_Udyam_Digital_7Slides.pptx

7-Slide Deck (PDF Format):
Airtel_SYNC_Udyam_Digital_7Slides.pdf

Comprehensive Submission Doc:
COMPREHENSIVE_SUBMISSION_DOC.md & privacy_note.md

Planted Traps Testing Key:
All 12 Specimen applicants evaluated; planted traps caught 100%